

# Varun Srivastava

408-896-0580 | [varun.neal@berkeley.edu](mailto:varun.neal@berkeley.edu) | [linkedin.com/in/varun-n-sri/](https://www.linkedin.com/in/varun-n-sri/) | [varunneal.github.io](https://varunneal.github.io)

## EDUCATION

**University of California, Berkeley**

*B.A. Computer Science, Mathematics*

Graduated Dec 2023

GPA: 3.9/4

## EXPERIENCE

**Machine Learning Engineer**

*AdMarketplace*

August 2024 – Present

New York, NY

- Designing and training ML time-series models for the firm's algorithmic ad bidding solution, optimizing millions of dollars in daily ad impressions.
- Engineering generative AI and LLM integrations to enhance search and recommendations across a catalog of millions of products for hundreds of publishers.
- Developing dashboards to evaluate ML model performance, collaborating cross-functionally with business and engineering teams to drive improvements.

**Founding Engineer**

*Stork Oracle Network*

August 2022 – December 2022

San Francisco, CA

- Designed a high-speed and reliable WebSocket API for live crypto data. The pricefeed served several notable paying DeFi clients, such as Vertex and Apex.
- Scaled the full-stack Python infrastructure up to meet high uptime and low latency requirements for platforms trading millions of dollars daily.

**Quantitative Analysis Intern**

*Dexterity Capital*

May 2022 – August 2022

Seattle, WA

- Implemented CMA-ES strategy in forecasting models, improving hyperparameter convergence by > 100x.
- Analyzed signals in cryptocurrency markets that became incorporated in the company's proprietary high frequency trading (HFT) market making algorithm, which trades billions of dollars in volume a day.

**Data Science Intern**

*First Republic Bank*

June 2020 – August 2020

San Francisco, CA

- Optimized the internal data orchestration (CI/CD) pipeline by designing an Apache Airflow pipeline with scheduled tasks and dependencies, decreasing end-to-end execution time of tasks by 50%.

## RESEARCH & PUBLICATIONS

Machine Learning & Vision Researcher | *Berkeley Visual Computing Lab*

December 2023 – May 2024

- Lee, J., **Srivastava, V.**, Jennings, N., Ng, R. "Theory of Human Tetrachromatic Color Experience." *ACM SIGGRAPH*, 2024. Honorable Mention for Technical Papers Awards, spotlighted at the annual conference.
- Researched the emergence of color vision in humans via neurological models, specifically in the case of esoteric conditions such as human tetrachromacy.

Vision & Neuroscience Researcher | *Na-Ji Biophysics Lab*

September 2019 – April 2020

- Implemented computer vision tools in Matlab for researchers to track the pupil size of rats in video data.
- Correlated terabytes of time-series data with pupil measurements to analyze how rodents react to shock stimulus.

## PROJECTS

**Spotify MCP** | [github.com/varunneal/spotify-mcp](https://github.com/varunneal/spotify-mcp)

- MCP Server integrating Spotify with LLMs (> 350 GitHub stars). Won the Anthropic MCP Hackathon.

**Semantic Finder** | [do-me.github.io/SemanticFinder](https://do-me.github.io/SemanticFinder)

- Developed a frontend-only semantic search site powered by LLMs (> 250 GitHub stars) and released a corresponding Chrome extension.

## TECHNICAL SKILLS

**Languages:** Python, C/C++, JavaScript, Go, Java, SQL, Matlab.

**Frameworks & development:** AWS, React, Jenkins, Docker, Bitbucket, Kubernetes Node, Django.

**Data & ML:** PyTorch, Huggingface, AWS Sagemaker, GPU computing (MLX, Cuda, Torch).